

Conditional Classification of Projection-Based Effective Theories:

Witness Records, Comparison Axes, and the Limits of Inevitability in Modal Triplet Theory

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Abstract

Irreversible records, ultraviolet control, event-like outcomes, and consistent matter sectors are genuine demands on physical theories. They do not, by themselves, force all viable theories into one mathematical universality class. The missing information is the mechanism: which states and operations are admitted, what is projected, what counts as recovery, which dynamics stabilizes records, how ultraviolet errors are controlled, how events are generated, and which global gauge and bundle data define the matter sector. This paper replaces an earlier inevitability claim by a conditional classification framework. A theory is represented by a seven-axis witness record covering operational measurement, descent and recovery, stability, ultraviolet control, event structure, gauge and matter data, and empirical validation. We prove five bounded results: noninjective descent has no left inverse; compact cores of open record regions have positive margins; transverse crossings are locally finite; representations of a specified compact $U(1)$ have integer weights; and multi-axis equivalence classes refine monotonically as comparison requirements are added. Explicit counterexamples show why operational irreversibility need not mean algebraic noninvertibility, low-energy predictivity does not select smooth spectral damping, countable outcomes do not create a causal set, and local anomaly cancellation or charge quantization does not determine the observed Standard Model. Collapse models, monitored circuits, effective field theory, asymptotic safety, loop quantum gravity, causal sets, noncommutative geometry, and Modal Triplet Theory are therefore compared only through completed witness rows. The result is a falsifiable taxonomy and release-ready comparison protocol, not a proof that the apparent plurality of physical theories is illusory.

Revision note for Version 3

- Supersedes.** Version 2, DOI 10.5281/zenodo.18268193.
- Reason.** Version 2 presented five broad “inevitability” theorems and concluded that all successful measurement, quantum-gravity, ultraviolet, and matter frameworks belong to a single coherent universality class. Those conclusions did not follow from the stated empirical requirements.
- Resolution.** Version 3 withdraws that classification, corrects the direction of the inverse used in projection arguments, distinguishes operational nonrecoverability from algebraic noninvertibility, replaces automatic basins and thresholds by typed stability hypotheses, separates renormalization from spectral damping, and restores the missing global gauge, anomaly-inflow, charge-normalization, and bundle data.

Retained content.

The paper retains a conditional witness-based taxonomy and the correctly scoped local implications proved from completed witness rows.

Remaining boundary.

No common ontology, ultraviolet completion, event source, observed matter spectrum, gravity theory, or global physical equivalence is inferred without a separate typed bridge and validation record.

This revision note records the release delta; it is not part of the abstract.

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1 The question that can actually be answered

Theories can resemble one another for several different reasons. Two microscopic models may have the same infrared fixed point. Two measurement models may produce the same outcome probabilities. Two compactifications may have the same low-energy gauge algebra. Two operators may have nearby compressed resolvents. None of these statements implies the others.

The word *universality* is therefore incomplete unless it names:

- (1) the objects being compared;
- (2) the maps that place them in one common space;
- (3) the topology, norm, or operational metric of comparison;
- (4) the parameter range and time horizon;
- (5) the observables whose agreement matters; and
- (6) the error and provenance of the comparison.

This distinction is especially important in Modal Triplet Theory (MTT). The current operator results establish controlled reduction inside a typed model and local comparison after retained spaces have been explicitly transported to a common Hilbert space [7, 8]. They do not prove that unrelated theories share one measurement dynamics, ultraviolet completion, event ontology, gauge group, or matter content.

The aim here is modest but useful. We ask:

What finite record must be supplied before two proposed theories can be said to belong to the same controlled class, and what follows from that record?

This converts a broad philosophical slogan into an auditable classification problem.

2 Desiderata are not mechanisms

2.1 Five empirical demands

For orientation, consider five demands often imposed on a candidate physical theory:

- **D_{rec}: records.** It reproduces stable macroscopic records and their observed statistics.
- **D_{loc}: locality.** It supplies the relevant relativistic locality or causal-compatibility property on the declared observables.
- **D_{pred}: predictivity.** It gives finite, controlled predictions in a stated energy and accuracy regime.
- **D_{event}: outcomes.** It explains or models individual registered outcomes when such outcomes are part of the intended ontology.
- **D_{matter}: matter.** It reproduces the observed gauge representations, anomaly structure, charge assignments, masses, mixings, and forbidden or suppressed processes to its declared tier.

These demands are not yet mathematical premises. For example, “irreversible record” does not say whether irreversibility means a noninjective map, absence of a completely positive recovery, thermodynamic cost, or practical inaccessibility of environmental degrees of freedom. Likewise, “finite predictivity” does not choose among renormalization, an effective cutoff, an ultraviolet fixed point, modular finiteness, a lattice continuum limit, or nonlocal spectral damping.

2.2 A claim ladder

Level	What is established	What is still not established
Vocabulary	A common list of words such as projection, basin, gap, event, or topology.	That the words denote the same mathematical objects.
Typed encoding	Each theory supplies objects and maps with compatible domains and codomains.	That the maps commute or preserve dynamics and observables.
Controlled comparison	A norm, region, transport map, and quantitative error are proved.	Global identity, arbitrary-time agreement, or physical equivalence.
Physical equivalence	Preparations, dynamics, observables, units, uncertainties, and interpretation are matched.	Microscopic identity or unique ontology.
Selection	One candidate is chosen by an independent source rule and passes held-out tests.	That every viable theory had to use the same mechanism.

Table 1: A comparison must not be promoted by skipping a level.

Version 2 moved directly from shared vocabulary and empirical desiderata to global physical classification. The remainder of this paper supplies the missing intermediate record.

3 The seven-axis witness record

Definition 3.1 (Classification witness). A *classification witness* for a model a is a tuple

$$W_a = (M_a, P_a, S_a, U_a, E_a, G_a, V_a; \pi_a),$$

with the following rows.

- A M_a , the operational-measurement row: state space, admissible preparations, instruments, outcomes, record channel, and operational metric.
- B P_a , the projection or descent row: source and effective spaces, descent map, declared recovery class, injectivity or recovery defect, and approximation error.
- C S_a , the stability row: dynamics, domain, metric or norm, record regions or metastable sets, margins, retention estimate, and time horizon.
- D U_a , the ultraviolet row: regulator or construction, renormalization and continuum prescription, symmetry identities, error or remainder, energy range, and status of ultraviolet completion.
- E E_a , the event row: path or history space, event functional, stopping/reset rule, ordering relation, local-finiteness or dwell-time condition, and outcome law.

- F G_a , the gauge-and-matter row: global gauge group, representations, charge normalization, spin and bundle data, anomaly functional including inflow or counterterms, and operator-selection rule.
- G V_a , the validation row: preparation and observable maps, unit conventions, matching scale, uncertainty budget, comparison data, and held-out tests.

The provenance row π_a contains source identifiers, versions, hashes, proof status, and known open obligations.

The rows deliberately overlap at their interfaces. An outcome law belongs to the measurement row, while the event row says how individual events are generated or encoded. A spectral gap belongs to the stability or operator certificate, while ultraviolet control must explain what happens across scales. The validation row prevents formal similarity from being mistaken for empirical agreement.

Definition 3.2 (Complete comparison on an axis set). Let $I \subseteq \{M, P, S, U, E, G, V\}$. A comparison of models a and b is *complete on I* when every row in I supplies:

- (i) explicit transports to common comparison objects;
- (ii) a declared exact relation or quantitative tolerance;
- (iii) all hypotheses needed for that relation;
- (iv) an error or obstruction certificate; and
- (v) source provenance.

Blank rows are not failures of a theory. They are failures of the proposed comparison. This is the central bookkeeping correction.

4 Projection, recovery, and operational irreversibility

4.1 Three notions that must not be conflated

Let X be a source state space, Y an effective state space, and $P : X \rightarrow Y$ a descent map.

- (1) P is *algebraically noninjective* if two source states have the same image.
- (2) P is *operationally nonrecoverable* relative to a class $\mathfrak{D}$ if no allowed decoder in $\mathfrak{D}$ recovers all source states.
- (3) A process is *thermodynamically or practically irreversible* if reversal lies outside a declared resource or control regime.

Only the first is a property of the set map alone.

Proposition 4.1 (Noninjective descent has no left inverse). *If $P : X \rightarrow Y$ is noninjective, there is no map $L : Y \rightarrow X$ satisfying*

$$L \circ P = \text{id}_X.$$

Proof. Choose $x_1 \neq x_2$ with $P(x_1) = P(x_2)$. If a left inverse existed, then

$$x_1 = L(P(x_1)) = L(P(x_2)) = x_2,$$

a contradiction. $\square$

Remark 4.2 (Left versus right). The obstructed object in theorem 4.1 is a *left* inverse of P . A right inverse $R : Y \rightarrow X$, satisfying $P \circ R = \text{id}_Y$, is a section and may exist for a surjective but noninjective map. Noninjectivity therefore proves loss of unique recovery, not absence of every section.

4.2 Operational nonrecoverability can occur without algebraic loss

Quantum operations are naturally described by completely positive maps and instruments [4]. A microscopic unitary dilation may coexist with a reduced, operationally irreversible channel [15]. The following elementary example blocks the old necessity argument.

Example 4.3 (Invertible linear map with no physical recovery). On d -dimensional density matrices let

$$\Phi_p(\rho) = p\rho + (1-p)\frac{I_d}{d}, \quad 0 < p < 1.$$

On the affine space of trace-one Hermitian matrices, Φ_p has the linear inverse

$$\Phi_p^{-1}(\sigma) = \frac{1}{p}\sigma - \frac{1-p}{pd}I_d.$$

That inverse is not a quantum channel. Indeed, for orthogonal pure states ρ, σ , trace distance contracts from 1 to p . If a completely positive trace-preserving map D recovered both states, contractivity would give

$$1 = D_{\text{tr}}(\rho, \sigma) = D_{\text{tr}}(D\Phi_p(\rho), D\Phi_p(\sigma)) \leq D_{\text{tr}}(\Phi_p(\rho), \Phi_p(\sigma)) = p,$$

which is impossible.

Thus “no admissible inverse” is often the correct operational statement. It depends on the recovery class. It is not equivalent to noninjectivity, and neither statement follows from the bare observation that records look irreversible. Objective-collapse models, for example, introduce explicit modified dynamics rather than deriving it from a general projection theorem [5]. The current MTT measurement paper likewise treats measurement as an ordinary physical interaction plus an explicitly typed completion and record process [9].

Definition 4.4 (Recovery defect). For a source set $K \subseteq X$, metric d_X , and allowed decoder class $\mathfrak{D}$, define

$$\delta_{\text{rec}}(P; K, \mathfrak{D}) = \inf_{D \in \mathfrak{D}} \sup_{x \in K} d_X(D(Px), x).$$

This scalar is meaningful only after K , d_X , and $\mathfrak{D}$ are fixed. It distinguishes exact information loss from restricted or approximate recovery.

5 Record stability is not automatically a dynamical basin

5.1 Record regions and positive margins

Let (Y, d) be an effective state space and $r : Y \rightarrow \mathcal{R}$ a record label. For a label α , write $B_\alpha = r^{-1}(\alpha)$. If B_α is open, nearby states share the same label. This is a kinematic robustness statement. A dynamical basin additionally requires a specified evolution and an attraction or retention property.

Proposition 5.1 (Compact record cores have positive margins). *Let B_α be open in a metric space Y , and let $K_\alpha \subset B_\alpha$ be compact. Then*

$$m_\alpha = \text{dist}(K_\alpha, Y \setminus B_\alpha) > 0,$$

unless $Y \setminus B_\alpha$ is empty, in which case the margin is unbounded.

Proof. The complement $Y \setminus B_\alpha$ is closed. The continuous function $y \mapsto \text{dist}(y, Y \setminus B_\alpha)$ is strictly positive on K_α . Compactness gives a positive minimum. $\square$

The proposition explains exactly when a finite stability margin is justified. It does not prove that every record region is open, that a useful compact core exists, or that dynamics preserves it.

Definition 5.2 (Finite-horizon record certificate). A finite-horizon record certificate consists of a propagator or channel family $\{\Phi_t\}_{0 \leq t \leq T}$, a record core K_α , a record region B_α , a metric d , and a bound

$$\inf_{y \in K_\alpha} \Pr_y[\Phi_t(y) \in B_\alpha \text{ for all } 0 \leq t \leq T] \geq 1 - \eta_\alpha.$$

For deterministic dynamics the probability is omitted.

Attraction, metastability, redundancy, superselection, and error correction can all stabilize records, but they are different mechanisms. A sharp “knee” in a control parameter is also not automatic. The smooth response

$$p(a) = \frac{1}{1 + e^{-a}}$$

changes between two record probabilities without a singular threshold. A phase transition or first-exit boundary must be proved in the chosen model. Measurement-induced entanglement transitions are genuine universality results for specified monitored many-body models, not a theorem about all measurement dynamics [14].

6 Ultraviolet control is a family of mechanisms

6.1 The ultraviolet row

A usable ultraviolet certificate must state at least:

- (1) the bare or regulated objects;
- (2) the renormalization or continuum-limit prescription;
- (3) the symmetry or Ward identities maintained or recovered;
- (4) the norm and observable class in which convergence is claimed;
- (5) the energy range and truncation error; and

- (6) whether the construction is an effective theory, a perturbative completion, or a nonperturbative completion.

Wilsonian coarse-graining and effective Lagrangians [17, 12], an asymptotic fixed point [13], modular finiteness, a discrete continuum limit, and an entire spectral form factor are not the same certificate. They may share low-energy predictions without sharing ultraviolet operators.

Proposition 6.1 (Low-energy agreement does not select spectral damping). *Fix $m > 0$, an accessible momentum bound $E > 0$, and $\Lambda > 0$. Define*

$$G_0(p) = \frac{1}{p^2 + m^2}, \quad G_\Lambda(p) = \frac{e^{-p^2/\Lambda^2}}{p^2 + m^2}.$$

Then for $|p| \leq E$,

$$|G_0(p) - G_\Lambda(p)| \leq \frac{E^2}{\Lambda^2 m^2}.$$

Consequently, low-energy agreement to any fixed tolerance does not determine whether the ultraviolet kernel contains smooth exponential damping.

Proof. Use $0 \leq 1 - e^{-x} \leq x$ for $x \geq 0$:

$$|G_0 - G_\Lambda| = \frac{1 - e^{-p^2/\Lambda^2}}{p^2 + m^2} \leq \frac{p^2}{\Lambda^2(p^2 + m^2)} \leq \frac{E^2}{\Lambda^2 m^2}.$$

□

The example does not claim that both kernels define the same complete theory. It proves the narrower logical point: finite predictivity on a bounded range cannot force one ultraviolet mechanism. It also shows why a four-dimensional smooth UV filter cannot be inferred merely from suppression of an internal operator sector.

7 When events are actually discrete

An outcome label, a quantum jump, a first-passage time, a detector click, and an element of a causal set are different objects. Their discreteness arises from different assumptions.

Theorem 7.1 (Transverse crossings are finite on compact time intervals). *Let $f \in C^1([0, T], \mathbb{R})$. If*

$$f(t) = 0 \implies f'(t) \neq 0,$$

then the zero set of f in $[0, T]$ is finite.

Proof. Every zero is isolated by the inverse-function theorem. If there were infinitely many zeros in the compact interval, they would possess an accumulation point t_* . Continuity gives $f(t_*) = 0$. But $f'(t_*) \neq 0$ makes t_* an isolated zero, contradicting accumulation. □

This theorem supports an event interpretation for smooth, transverse first-exit models on finite horizons. Without the hypothesis, zeros may accumulate; stochastic paths may revisit a boundary infinitely often; reset dynamics may exhibit Zeno behavior; and an outcome instrument may be discrete without any underlying boundary crossing.

A causal set requires more still: a partial order and local finiteness of order intervals. Those are defining structural data in the causal-set program [2]. A countable list of basin transitions does not by itself supply either property. Conversely, discrete spacetime proposals cannot be rejected merely by saying that a lattice violates Lorentz symmetry; the actual construction and continuum approximation must be examined.

8 Gauge and matter claims require global data

8.1 The gauge-and-matter row

At minimum, a matter certificate must specify:

$$\mathbf{G} = (G_{\text{global}}, \{\rho_i\}, \mathcal{H}_{\text{ferm}}, \mathfrak{s}, \mathcal{L}, \mathcal{A}_{\text{tot}}, \mathcal{O}_{\text{allowed}}, \nu).$$

Here G_{global} is the global gauge group rather than only its Lie algebra; ρ_i are representations; $\mathfrak{s}$ records spin or Spin^c data; $\mathcal{L}$ records bundles and characteristic classes; $\mathcal{A}_{\text{tot}}$ is the complete anomaly including inflow and counterterms; $\mathcal{O}_{\text{allowed}}$ is the operator-selection rule; and ν fixes charge normalization.

8.2 What a compact circle group does prove

Proposition 8.1 (Weights of a specified compact circle). *Every finite-dimensional continuous unitary representation of $U(1)$ is a direct sum of characters*

$$z \mapsto z^n, \quad n \in \mathbb{Z}.$$

Thus charges are integer weights after a generator and normalization have been fixed.

Proof. The commuting unitary matrices in the image can be simultaneously diagonalized. Each one-dimensional continuous character of $U(1)$ has the form $e^{i\theta} \mapsto e^{in\theta}$ with $n \in \mathbb{Z}$. $\square$

This proposition does not derive the observed hypercharges. Fractional numerical values depend on normalization and on the global quotient connecting the $U(1)$ factor to nonabelian centers. Replacing $U(1)$ by the additive group $\mathbb{R}$ also removes the integer-character conclusion. The common-circle idea in MTT is therefore potentially useful only after the same physical line, connection, and global group action are identified.

8.3 Anomalies and operator selection

Gauge consistency constrains the *total* anomaly. It need not require the bare fermion polynomial to vanish term by term: Green–Schwarz inflow is a standard counterexample [6]. A theorem must state the dimension, chiral content, regularization, local and global anomalies, inflow, and boundary conditions.

Proposition 8.2 (Conditional bundle obstruction). *Suppose fields ψ_i are sections of associated bundles E_i , and a candidate local monomial with constant scalar coefficient transforms as a section of a product bundle $E_{\mathcal{O}}$. If $E_{\mathcal{O}}$ admits no gauge-invariant identification with the scalar-density bundle, that monomial cannot define a global gauge-invariant action term with that coefficient. Conversely, such an identification removes this obstruction but does not force the term to occur or fix its coefficient.*

Proof. A global action density must be a gauge-invariant scalar density. Without an equivariant identification of $E_{\mathcal{O}}$ with that target, local expressions fail to glue to a global scalar. If an identification exists, the gluing obstruction is absent, but dynamics and additional symmetries still decide whether the coefficient vanishes. $\square$

This is the correct scope for topology-only exclusion. It cannot be promoted to the statement that all dangerous baryon- or lepton-number violating operators are absent in every admissible theory. Such operators are systematically present in effective-field-theory classifications unless additional symmetries or bundle rules exclude them [16]. Nor do anomaly freedom, chirality, and charge quantization alone prove that the Standard Model is the unique or near-minimal solution.

9 The mathematics of a multi-axis classification

9.1 Exact classes

Let $\mathfrak{T}$ be a set of typed model records. For each axis $k \in K$, let

$$I_k : \mathfrak{T} \rightarrow Q_k$$

be the invariant extracted from that row, and let $\simeq_k$ be an equivalence relation on Q_k .

Definition 9.1 (Axis-set equivalence). For $S \subseteq K$, define

$$a \sim_S b \iff I_k(a) \simeq_k I_k(b) \text{ for every } k \in S.$$

Theorem 9.2 (Intersection and refinement of classification axes). *For every $S \subseteq K$, the relation $\sim_S$ is an equivalence relation. If $S \subseteq T \subseteq K$, then every $\sim_T$ -class is contained in a $\sim_S$ -class.*

Proof. The relation $\sim_S$ is the intersection of the equivalence relations induced by the maps I_k , so it is reflexive, symmetric, and transitive. If $a \sim_T b$, then the defining conditions hold for every $k \in T$, hence for every $k \in S$. $\square$

The theorem is elementary, but it carries the main methodological message: adding measurement, ultraviolet, matter, or validation rows can only refine a class. Agreement on one row cannot establish agreement on a stronger set.

Proposition 9.3 (No inference between independent axes). *Let $Q_1, \dots, Q_n$ each contain at least two elements, and take $\mathfrak{T} = Q_1 \times \dots \times Q_n$ with I_k the coordinate projection. For any proper subset $S \subsetneq \{1, \dots, n\}$, there exist $a, b \in \mathfrak{T}$ such that $a \sim_S b$ but $I_j(a) \neq I_j(b)$ for some $j \notin S$.*

Proof. Choose a and b equal in every coordinate in S , and choose distinct values in one coordinate $j \notin S$. $\square$

Physical axes may be coupled by nontrivial theorems. The proposition says that the coupling must be proved; it is not supplied by the vocabulary.

9.2 Approximate classes

A metric tolerance such as

$$d_k(I_k(a), I_k(b)) \leq \varepsilon_k$$

is generally not transitive. It defines a comparison neighborhood or graph, not automatically an equivalence class. To speak of a genuine universality class one needs an exact equivalence, a quotient construction, a common limit, or a controlled transitive closure whose accumulated error remains acceptable. The local coherent-sector comparison paper proves the relevant resolvent bound and explicitly tracks chainwise error accumulation [8]. It does not license unlimited chains.

Definition 9.4 (Conditional coherent comparison class). A family $\mathfrak{C} \subseteq \mathfrak{T}$ is a *conditional coherent comparison class* on axis set S when:

- (i) every model has a complete witness on S ;
- (ii) the transport maps and comparison relations are fixed before testing;
- (iii) exact equivalence or a uniformly controlled common-limit theorem is proved on each axis in S ; and
- (iv) the validation row states what physical agreement follows.

This definition deliberately avoids the word “inevitable.” Membership is a verified property of supplied records.

10 What the major external frameworks actually contribute

The comparison below records native structures, not verdicts on entire research programs.

Framework	Native contribution relevant here	What does not follow automatically
Operational quantum theory	Instruments, completely positive operations, outcome probabilities, and sequential measurement [4].	A unique ontology, objective event source, or irreversible microscopic law.
Objective collapse	Explicit stochastic modification producing localization and macroscopic classicality [5].	Derivation from generic record stability or equivalence to coarse-graining.
Monitored circuits	Model-specific entanglement phases and critical behavior under measurement [14].	A universal solution of measurement or one class for all instruments.
EFT and RG	Controlled low-energy expansions, coarse-graining, running couplings, and universality near fixed points [17, 12].	A unique UV completion, literal irreversibility of every beta-function ODE, or smooth spectral damping.
Asymptotic safety	A candidate non-Gaussian fixed-point mechanism formulated through scale-dependent effective actions [13].	Equality to string modular control or an entire form-factor kernel.
Loop quantum gravity	Background-independent quantum geometry and a mathematically specific kinematical representation [1].	An MTT projector, measurement completion, or full low-energy equivalence without an explicit bridge.
Causal sets	A locally finite partial order proposed as microscopic spacetime structure [2].	Derivation from generic basin crossings or failure of Lorentzian approximation.
Noncommutative geometry	A spectral triple and spectral action once the algebra, Hilbert space, and Dirac operator are supplied [3].	Selection of that triple from a generic coherent sector or uniqueness of the observed Standard Model.

Table 2: A framework should be compared through its own typed objects.

The table reverses the old “partial shadows” rhetoric. Similarity is now a question to be certified, not an interpretation imposed in advance.

11 The present MTT position

MTT currently supplies several distinct layers:

- (1) The Foundations paper defines the closure/admissibility program and its typed status boundaries [10].
- (2) The Projection–Admissibility paper distinguishes descent, recoverability, and structural constraints on an effective description [11].

- (3) The coherent-sector reduction paper proves a domain-explicit single-model Feshbach and projector certificate, including the special case in which an exact spectral projector makes the off-diagonal correction vanish [7].
- (4) The local robustness paper transports retained resolvents to one reference Hilbert space and proves bounded local comparison, while separating projector, operator, dynamical, observable, and physical equivalence [8].
- (5) The measurement paper separates physical disturbance, outcome completion, record stabilization, and the still-open source of a general outcome law [9].

These are meaningful advances, but they do not close all seven witness rows for MTT or for any external framework. In particular:

- a selected local projector is not a universal measurement instrument;
- an internal spectral filter is not by itself a four-dimensional UV completion;
- a rank or bundle flag is not yet the observed matter spectrum;
- a finite event encoding is not automatically a causal set;
- local resolvent closeness is not physical equivalence; and
- one successful MTT realization would not prove that all successful theories had to use the same microscopic mechanism.

The phrase *coherent universality* may still be used, but only with an axis set and certificate. For example, “local norm-resolvent coherent universality on Ω ” is precise. “All viable physics belongs to one coherent universality class” is not presently a theorem.

12 A practical comparison protocol

For a proposed comparison between MTT and another framework:

- Step 1.** Freeze the exact source versions and their provenance hashes.
- Step 2.** Choose the axis set S . Do not add untested axes in the conclusion.
- Step 3.** Fill both witness records, including domains, norms, units, and time or energy ranges.
- Step 4.** Construct explicit transports to common objects.
- Step 5.** Prove commutation, convergence, or a quantitative error bound.
- Step 6.** Map preparations, states, observables, and predictions.
- Step 7.** Reserve at least one independent or held-out test.
- Step 8.** Record failed rows and breakdown surfaces as part of the result.

12.1 Minimum machine-readable payload

A computational record should include:

```
model_id, source_hashes, axis_set,  
state_space, domains, transports, comparison_norm,  
parameter_region, time_or_energy_horizon,  
error_bounds, observable_map, unit_map,  
status, failed_rows, verifier, artifact_hashes
```

The payload does not replace the proof. It prevents a later comparison from silently changing its objects or standard.

13 Completion and falsifiability

The framework can fail in informative ways.

- (F1) A claimed descent map is injective, while the argument requires information loss.
- (F2) The recovery defect vanishes in the declared physical decoder class.
- (F3) Record regions have zero margin or fail finite-horizon retention.
- (F4) The ultraviolet row lacks a continuum or error certificate.
- (F5) event times accumulate, or no causal order is supplied.
- (F6) the global gauge group, anomaly total, or bundle product contradicts the proposed matter claim.
- (F7) transported observables disagree beyond the declared uncertainty.

Conversely, completing all rows would justify a strong result: not that two theories are ontologically identical, but that they are equivalent on a declared physical interface to a certified accuracy.

14 Conclusion

The strongest correction in this paper is also the simplest. Empirical demands constrain theories, but they do not name the mathematical mechanism satisfying them. Irreversible records do not alone force a noninjective projection. Stable records do not uniquely force attractor basins or sharp knees. Finite predictivity does not select smooth spectral damping. Event-like outcomes do not automatically define a causal set. Gauge consistency and compact-circle weights do not uniquely determine the observed matter sector.

What survives is a rigorous and reusable classification method. The seven-axis witness record shows exactly what is common, what has merely been translated, what is quantitatively controlled, and what remains open. The exact classification theorem says that adding axes refines rather than preserves a class; the counterexamples show why cross-axis conclusions require their own bridges.

For MTT, this is not a retreat from comparison. It is what makes comparison credible. The released operator results now occupy their correct local layer, while measurement, ultraviolet completion, event structure, matter selection, and empirical equivalence remain independently testable rows. A future theorem may couple some of those rows. Until it does, the correct claim is conditional classification, not inevitability.

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